

$$C_{\phi 1 \phi 2} g_2^4 L F_{3,0}[m_2] + \frac{1}{3} C_{\phi 1 \phi 2} g_2^4 L F_{4,-1}[m_2] - \frac{16}{45} C_{\phi 1 \phi 2} g_2^4 L F_{5,-2}[m_2] + \\ C_{\phi 1 \phi 2} g_2^2 (9 \overline{y_e^{PR}} y_e^{PR} + 4 \sum_p g_2^2) L F_{3,0}[m_l^P] - \frac{1}{24} C_{\phi 1 \phi 2} g_2^2 (6 \overline{y_e^{PR}} y_e^{PR} + 5 \sum_p g_2^2) L F_{4,-1}[m_l^P] + \\ C_{\phi 1 \phi 2} g_2^4 L F_{5,-2}[m_l^P] + \frac{1}{12} C_{\phi 1 \phi 2} g_2^2 (9 \overline{y_d^{PR}} y_d^{PR} - 9 \overline{y_u^{PR}} y_u^{PR} + 4 \sum_p g_2^2) L F_{3,0}[m_q^P] - \\ C_{\phi 1 \phi 2} g_2^2 (6 \overline{y_d^{PR}} y_d^{PR} - 6 \overline{y_u^{PR}} y_u^{PR} + 5 \sum_p g_2^2) L F_{4,-1}[m_q^P] + \frac{4}{15} \sum_p C_{\phi 1 \phi 2} g_2^4 L F_{5,-2}[m_q^P] + \\ C_{\phi 1 \phi 2} g_2^4 L F_{3,0}[\tilde{\mu}] + \frac{1}{6} C_{\phi 1 \phi 2} g_2^4 L F_{4,-1}[\tilde{\mu}] - \frac{8}{45} C_{\phi 1 \phi 2} g_2^4 L F_{5,-2}[\tilde{\mu}] - m_1 \tilde{\mu} g_1^4 L F_{2,2,-1}[m_1, \tilde{\mu}] - \\ C_{\phi 1 \phi 2} g_2^4 L F_{2,1,0}[m_2, \tilde{\mu}] - \frac{12}{12} g_2^4 (13 C_{\phi 1 \phi 2} + 36 m_2 \tilde{\mu}) L F_{2,2,-1}[m_2, \tilde{\mu}] + \\ C_{\phi 1 \phi 2} g_2^4 (-3 m_2 C_{\phi 1 \phi 2} + 2 \tilde{\mu} (C_{\phi 1^2} + 3 C_{\phi 2^2})) L F_{2,2,0}[m_2, \tilde{\mu}] + \\ C_{\phi 1 \phi 2} g_2^4 L F_{3,1,-1}[m_2, \tilde{\mu}] + \frac{2}{3} C_{\phi 1 \phi 2} g_2^4 L F_{3,2,-2}[m_2, \tilde{\mu}] - \\ C_{\phi 1 \phi 2} g_2^4 (C_{\phi 1^2} + 3 C_{\phi 2^2}) L F_{3,2,-1}[m_2, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_d^{PR}} a_d^{PR} L F_{2,1,0}[m_d^r, m_q^P] + \\ (C_{\phi 1 \phi 2} \overline{a_d^{PR}} a_d^{PR} + \tilde{\mu} \overline{y_d^{PR}} (a_d^{PR} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_d^{PR})) L F_{2,2,0}[m_d^r, m_q^P] + \\ C_{\phi 1 \phi 2} \overline{y_e^{PR}} a_e^{PR} L F_{2,1,0}[m_e^r, m_l^P] + \\ (C_{\phi 1 \phi 2} \overline{a_e^{PR}} a_e^{PR} + \tilde{\mu} \overline{y_e^{PR}} (a_e^{PR} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_e^{PR})) L F_{2,2,0}[m_e^r, m_l^P] - \\ C_{\phi 1 \phi 2} \overline{y_e^{PR}} a_e^{PR} L F_{2,1,0}[m_l^P, m_e^r] + \frac{1}{4} \tilde{\mu} \overline{y_e^{PR}} a_e^{PR} (g_1^2 + g_2^2) L F_{3,1,-1}[m_l^P, m_e^r] + \\ C_{\phi 1 \phi 2} \overline{a_e^{PR}} a_e^{PR} (-g_1^2 + g_2^2) - 2 \tilde{\mu} g_1^2 \overline{y_e^{PR}} (a_e^{PR} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_e^{PR})) L F_{3,1,0}[m_l^P, m_e^r] - \\ (C_{\phi 1 \phi 2} \overline{a_e^{PR}} a_e^{PR} + \tilde{\mu} \overline{y_e^{PR}} (a_e^{PR} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_e^{PR})) L F_{3,2,-1}[m_l^P, m_e^r] + \\ C_{\phi 1 \phi 2} \overline{a_e^{PR}} a_e^{PR} (3 g_1^2 + g_2^2) + \tilde{\mu} \overline{y_e^{PR}} (a_e^{PR} (9 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) + g_2^2 (5 C_{\phi 1^2} + 9 C_{\phi 2^2}))) + \\ C_{\phi 1 \phi 2} \tilde{\mu} y_e^{PR} (9 g_1^2 + 5 g_2^2)) L F_{4,1,-1}[m_l^P, m_e^r] + \frac{1}{6} (-3 C_{\phi 1 \phi 2} g_2^2 \overline{a_e^{PR}} a_e^{PR} + \\ \overline{y_e^{PR}} (a_e^{PR} (-3 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) - 2 g_2^2 (2 C_{\phi 1^2} + 3 C_{\phi 2^2})) - C_{\phi 1 \phi 2} \tilde{\mu} y_e^{PR} (3 g_1^2 + 4 g_2^2))) \\ (-2 [m_l^P, m_e^r] - \frac{1}{2} C_{\phi 1 \phi 2} \overline{y_e^{PR}} \overline{y_e^{st}} y_e^{Pt} y_e^{Sr} L F_{2,1,0}[m_l^P, m_l^S] + \\ \overline{y_e^{PR}} \overline{y_e^{st}} y_e^{Pt} y_e^{Sr} L F_{3,1,-1}[m_l^P, m_l^S] - \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_d^{PR}} a_d^{PR} L F_{2,1,0}[m_q^P, m_d^r] + \\ C_{\phi 1 \phi 2} \overline{a_d^{PR}} a_d^{PR} (g_1^2 + g_2^2) L F_{3,1,-1}[m_q^P, m_d^r] + \frac{1}{4} \\ C_{\phi 1 \phi 2} \overline{a_d^{PR}} a_d^{PR} (g_1^2 + 3 g_2^2) - 2 \tilde{\mu} g_1^2 \overline{y_d^{PR}} (a_d^{PR} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_d^{PR})) L F_{3,1,0}[m_q^P, m_d^r] - \\ (C_{\phi 1 \phi 2} \overline{a_d^{PR}} a_d^{PR} + \tilde{\mu} \overline{y_d^{PR}} (a_d^{PR} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_d^{PR})) L F_{3,2,-1}[m_q^P, m_d^r] + \\ C_{\phi 1 \phi 2} \overline{a_d^{PR}} a_d^{PR} (-g_1^2 + g_2^2) + \\ C_{\phi 1 \phi 2} \overline{a_d^{PR}} (a_d^{PR} (5 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) + g_2^2 (5 C_{\phi 1^2} + 9 C_{\phi 2^2})) + 5 C_{\phi 1 \phi 2} \tilde{\mu} y_d^{PR} (g_1^2 + g_2^2))) \\ (-1 [m_q^P, m_d^r] + \frac{1}{2} (-3 C_{\phi 1 \phi 2} g_2^2 \overline{a_d^{PR}} a_d^{PR} + \\ \overline{y_d^{PR}} (a_d^{PR} (-3 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) - 2 g_2^2 (2 C_{\phi 1^2} + 3 C_{\phi 2^2})) - C_{\phi 1 \phi 2} \tilde{\mu} y_d^{PR} (3 g_1^2 + 4 g_2^2))) \\ (-2 [m_q^P, m_d^r] + \frac{3}{4} C_{\phi 1 \phi 2} \overline{y_d^{PR}} y_d^{Sr} (-2 \overline{y_d^{st}} y_d^{Pt} + \overline{y_u^{st}} y_u^{Pt}) L F_{2,1,0}[m_q^P, m_q^S] + \\ \overline{y_d^{PR}} y_d^{Sr} (2 \overline{y_d^{st}} y_d^{Pt} - \overline{y_u^{st}} y_u^{Pt}) L F_{3,1,-1}[m_q^P, m_q^S] + \\ C_{\phi 1 \phi 2} \overline{a_u^{PR}} a_u^{PR} (g_1^2 - 3 g_2^2) L F_{2,1,0}[m_q^P, m_u^r] + \frac{1}{8} (C_{\phi 1 \phi 2} \overline{a_u^{PR}} a_u^{PR} (g_1^2 - g_2^2) + \tilde{\mu} \overline{y_u^{PR}} \\ a_u^{PR} (g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) - g_2^2 (C_{\phi 1^2} - 3 C_{\phi 2^2})) + C_{\phi 1 \phi 2} \tilde{\mu} y_u^{PR} (g_1^2 + 3 g_2^2))) L F_{2,2,0}[m_q^P, m_u^r] - \\ (C_{\phi 1 \phi 2} \overline{a_u^{PR}} a_u^{PR} + \tilde{\mu} \overline{y_u^{PR}} (a_u^{PR} (C_{\phi 1^2} + 3 C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_u^{PR})) L F_{3,1,0}[m_q^P, m_u^r] + \\ C_{\phi 1 \phi 2} \overline{a_u^{PR}} a_u^{PR} + \tilde{\mu} \overline{y_u^{PR}} (a_u^{PR} (C_{\phi 1^2} + 3 C_{\phi 2^2})) L F_{3,2,-1}[m_q^P, m_u^r] + \\ (C_{\phi 1 \phi 2} \overline{a_u^{PR}} a_u^{PR} + \tilde{\mu} \overline{y_u^{PR}} (a_u^{PR} (C_{\phi 1^2} + 3 C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_u^{PR})) L F_{4,1,-1}[m_q^P, m_u^r] + \\ \overline{y_d^{PR}} y_d^{Sr} \overline{y_u^{st}} y_u^{Pt} L F_{2,1,0}[m_q^S, m_q^P] - \frac{3}{4} C_{\phi 1 \phi 2} \overline{y_d^{PR}} y_d^{Sr} \overline{y_u^{st}} y_u^{Pt} L F_{3,1,-1}[m_q^S, m_q^P] - \\ C_{\phi 1 \phi 2} \overline{a_u^{PR}} a_u^{PR} (7 g_1^2 + 3 g_2^2) L F_{2,1,0}[m_u^r, m_q^P] + \\ C_{\phi 1 \phi 2} \overline{a_u^{PR}} a_u^{PR} (g_1^2 + g_2^2) L F_{3,1,-1}[m_u^r, m_q^P] + \frac{1}{4} (-C_{\phi 1 \phi 2} \overline{a_u^{PR}} a_u^{PR} (7 g_1^2 + 4 g_2^2) + \\ \overline{y_u^{PR}} (a_u^{PR} (-7 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) - 2 g_2^2 (2 C_{\phi 1^2} + 3 C_{\phi 2^2})) - C_{\phi 1 \phi 2} \tilde{\mu} y_u^{PR} (4 g_1^2 + 3 g_2^2))) \\ (-1 [m_u^r, m_q^P] + \frac{1}{8} (C_{\phi 1 \phi 2} \overline{a_u^{PR}} a_u^{PR} (-g_1^2 + g_2^2) + \\ \overline{y_u^{PR}} (a_u^{PR} (-g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) + g_2^2 (C_{\phi 1^2} - 3 C_{\phi 2^2})) - C_{\phi 1 \phi 2} \tilde{\mu} y_u^{PR} (g_1^2 + 3 g_2^2))) \\ (-1 [m_u^r, m_q^P] + \frac{3}{4} (C_{\phi 1 \phi 2} \overline{a_u^{PR}} a_u^{PR} (5 g_1^2 + 4 g_2^$$